

BOUAFIA MOUAAD NAWFEL

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EDUCATION

University of Sciences and Technology Houari Boumediene (USTHB)

Bachelor of Science in Computer Science / Software Engineering Track

Algiers, Algeria

2022 – Present

EXPERIENCE & ENGINEERING PRACTICE

Software Engineer & Systems Architect

Algiers, Algeria

Independent Software Engineering Projects

2023 – Present

- Architected asynchronous high-throughput event streaming services utilizing **Java**, **Spring Boot**, and **Apache Kafka** with topic partitioning and consumer group scaling.
- Engineered modular microservices infrastructure integrating **Resilience4j** circuit breakers, dynamic **Eureka** service discovery, and **Spring Security JWT** authentication.
- Containerized cloud-native backend environments using **Docker** and orchestrated workloads via **Kubernetes** with automated deployment manifests.
- Designed RESTful endpoints and high-performance **gRPC** protocols linked with **PostgreSQL** relational storage and **Redis** in-memory caching.

KEY PROJECTS

Distributed Event Streaming Engine

Java, Spring Boot, Kafka, Docker

- Built a fault-tolerant log retention and message delivery system simulating dynamic worker pool rebalancing under high packet throughput.

High-Performance API Gateway Core

Go, gRPC, Redis, Prometheus

- Implemented a low-latency reverse proxy featuring distributed token bucket rate limiting and real-time telemetry extraction.

Enterprise Microservices Platform

Spring Boot, Security, PostgreSQL, K8s

- Developed an enterprise ecosystem with central configuration management, automated health probes, and Kubernetes cluster orchestration.

Real-Time Telemetry & Systems Management Portal

React, Tailwind CSS, Spring Boot, WebSocket

- Designed a full-stack dashboard rendering live metrics, node CPU/memory load graphs, and real-time system alerts over WebSocket connections.

PROGRAMMING & TECHNICAL SKILLS

Languages: Java, Go, Python, JavaScript/TypeScript, SQL, C/C++

Frameworks & Libraries: Spring Boot, Spring Security, Hibernate, React, Tailwind CSS, gRPC

Infrastructure & Tools: Microservices, Apache Kafka, Docker, Kubernetes, Redis, PostgreSQL, Git, Linux